Engineers decide. AI assembles and reasons over evidence. Rules enforce. The model never holds write authority.

INTAKE

- **Harness bar** — Injector, reset, scenarios
- **Three plants** — Northwater, Ironwood, Lumen
- **Sites API** — List, one site, compare
- **Scenario evidence** — Script records, full text
- **Notices and reports** — Operator, NCR, supplier

telemetry

DETERMINISTIC — code, unit tested

- **SPC windows** — Western Electric rules 1-4
- **Ticketing** — Alert opens a TKT
- **Intake** — Validate, dedupe, severity
- **Lifecycle** — Only legal transitions

channels

- **Impact and sites** — Scope, timing, and a plant compare. SC-5 lot reaches two plants

to orchestrator

READ-ONLY EVIDENCE

- **MES** — Runs, lots, recipes
- **QMS** — NCRs and CAPAs
- **PLM** — Configs, BOM, ECOs
- **ERP** — Lots and suppliers
- **CMMS** — PM and calibrations
- **Ticketing** — TKT records
- **Corpus and scenarios** — Procedures, bulletins, and script records
- **Chunk and embed** — Heading-aware chunks, versioned index
- **Hybrid retrieval** — Keyword plus vector, family and revision filters

read-only

JUDGMENT PATH

- **Orchestrator** — Workflow behind one interface
- **Investigation Agent** — Read-only tools, at most 10 calls
- **Evidence gate** — Gaps escalate. No cause asserted.
- **Resolution Agent** — Full record text. No tools.
- **Grounding validator** — Cited ids must be in the pack
- **Meridian console** — Sites through CAPA. Harness bar stays outside
- **Escalation gate** — No bare approve
- **Approval token** — Override needs a reason
- **Action service** — CAPA write. Not an agent tool.
- **Model gateway** — Route, retry, breaker, and a $2 per day hard stop
- **Azure OpenAI** — gpt-5.4-mini, one deployment
- **OpenAI** — Same gateway route
- **Replay** — Over cap, or local default
- **Mock reasoner** — No recording for this fault
- **Audit log** — Append-only. SPC, workflow, gateway.
- **Eval runner** — 24 cases. Hard gates on unsafe answers.
- **Red-team** — Injection doc and operator text
- **Registry** — Prompt, model, and index versions

LEGEND

- Deterministic rules. Code, unit tested.
- Probabilistic agents. No write tools.
- Human authority. Escalation blocks a bare approve.
- Read-only evidence, sites, and scenario records.
- Model gateway and the spend cap.
- Arrow: a handoff or a read.

MERIDIAN PRECISION SYSTEMS
Quality and Engineering Issue Resolution

Logical architecture
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